# SERVICE ORIENTED ARCHITECTURES

ACIT3855 – WINTER 2026

# AGENDA – LESSON 10

- Quick Review
- Quiz
- Topics:
  - Dashboard UI
  - CORS
- Lab 10 – CORS and Dashboard UI

# REVIEW

- Same Origin Policy
- Cross Origin Resource Sharing
- Web / front end technologies

# QUIZ

- Quiz is on the Learning Hub
- Open book, but do your own work
- You have <15 minutes to complete it

# COURSE SCHEDULE

| Week | Topics | Notes |
|---|---|---|
| 1 | • Services Based Architecture Overview<br>• RESTful APIs Review | Lab 1 |
| 2 | • Microservices Overview<br>• Edge Service | Lab 2, Quiz 1 |
| 3 | • Database Per Service<br>• Storage Service (SQLite) | Lab 3, Quiz 2 |
| 4 | • Logging, Debugging and Configuration<br>• Storage Service (MySQL) | Lab 4, Quiz 3 |
| 5 | • RESTful API Specification (OpenAPI)<br>• Processing Service | Lab 5, Quiz 4 |
| 6 | • Synchronous vs Asynchronous Communication<br>• Message Broker Setup and Messaging | Lab 6, Quiz 5 |
| 7 | • Event Sourcing<br>• Analysis Service | Lab 7 |
| 8 | **Midterm Exam** | |
| 9 | • Deployment Part 1 - Containerization (Docker/Docker Compose) | Lab 8, Quiz 6 |
| 10 | • Deployment Part 2 – Containerization (Docker Volumes and Bind Mounts) | Lab 9, Quiz 7 |
| 11 | **Computing Reading Week** | |
| 12 | • Dashboard UI and CORS | Lab 10, Quiz 8 |
| 13 | • Deployment Part 3 – Provisioning<br>• Issues and Technical Debt | Lab 11, Quiz 9 |
| 14 | • Deployment Part 4 – Load Balancing and Scaling<br>• Final Exam Preview | Lab 12, Quiz 10 |
| 15 | • Review Week | Lab 12 Demos, Assignment Due |
| 16 | **Final Exam** | |

# DASHBOARD USER INTERFACE

Dashboard UI

- Contain information such as stats, analytics, schedules, messages and much more
- Typically dynamic – changes as the system data changes

Single Page Application (SPA)

- Dynamically rewrites the page in the browser based on user input and/or requests to a backend
- Typically developed using a JavaScript framework like Angular or React
- Creates a more traditional desktop experience to the user as the page is not completely reloading after each action.

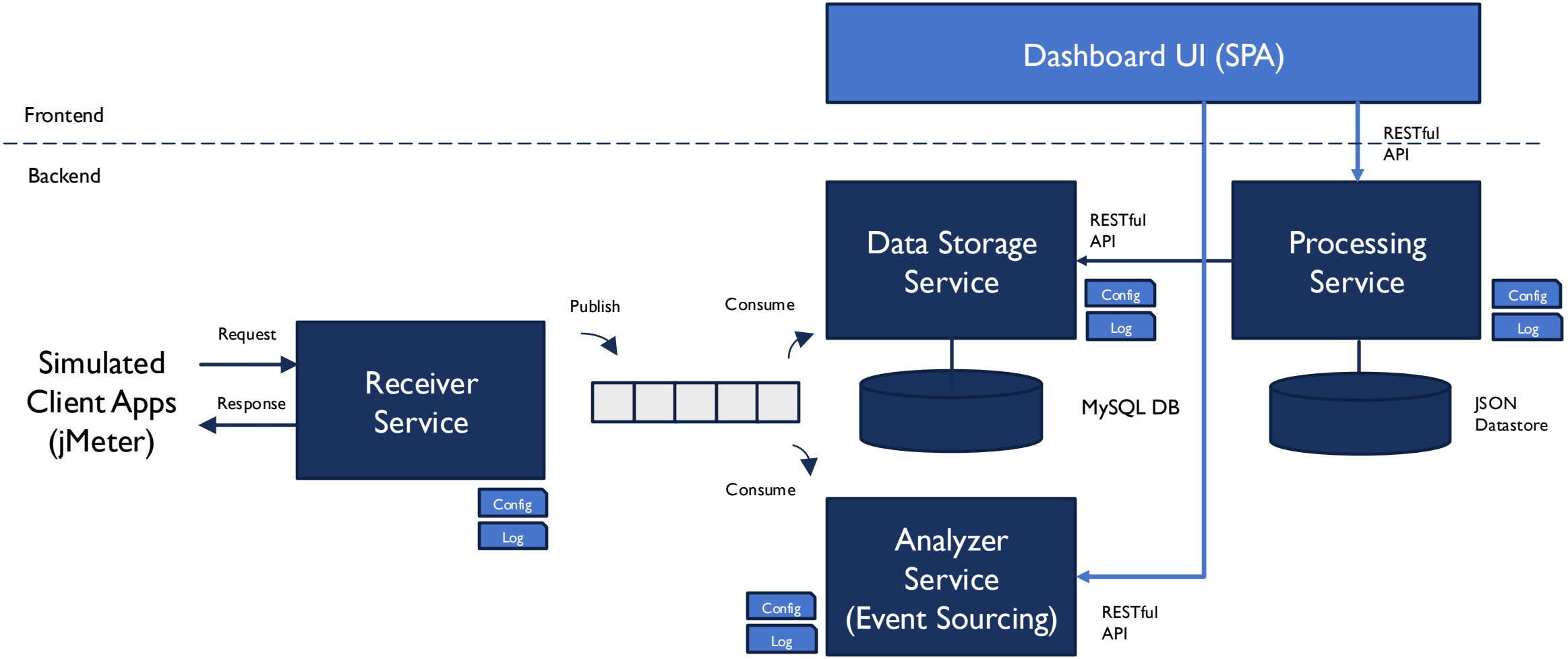
# SPA CHARACTERISTICS

- Web Server serves HTML/CSS/JavaScript pages to the user's browser
- JavaScript dynamically updates the HTML/CSS as the user performs actions and upon requests/responses to backend services
- Frontend – The SPA
- Backend – The RESTful services providing the GET/POST/PUT/DELETE methods to the front end
- Developers sometimes specialize in the Frontend or the Backend, or work across both as in Full-Stack Developers

# SOP AND CORS

- Often the server for your Frontend SPA and the server(s) for your Backend services are hosted on the same domain (i.e., *www.example.com*)
- But what if your Backend services are on a different domain (i.e., *www.example.com* for the Frontend and *api.example.com* for the Backend)?
  - The Browser has a restriction called SOP (Single Origin Policy). It will prevent a Frontend on one domain from calling Backend services on a different domain.
  - This can be particularly troublesome for Microservices where there are many services possible deployed on different domains
  - What are the solutions?
    - Proxy (Client Side) – Frontend has a proxy so that all backend services calls go to the same domain but are redirected to the correct domain
    - Reverse Proxy (Server Side) – Backend services are deployed behind a reverse proxy so they are all accessed from the same domain
    - Enable CORS (Cross-Origin Resource Sharing). on the backend services (i.e., tell the Frontend client it's okay to call this service)

# OUR SAMPLE APPLICATION – LAB 10



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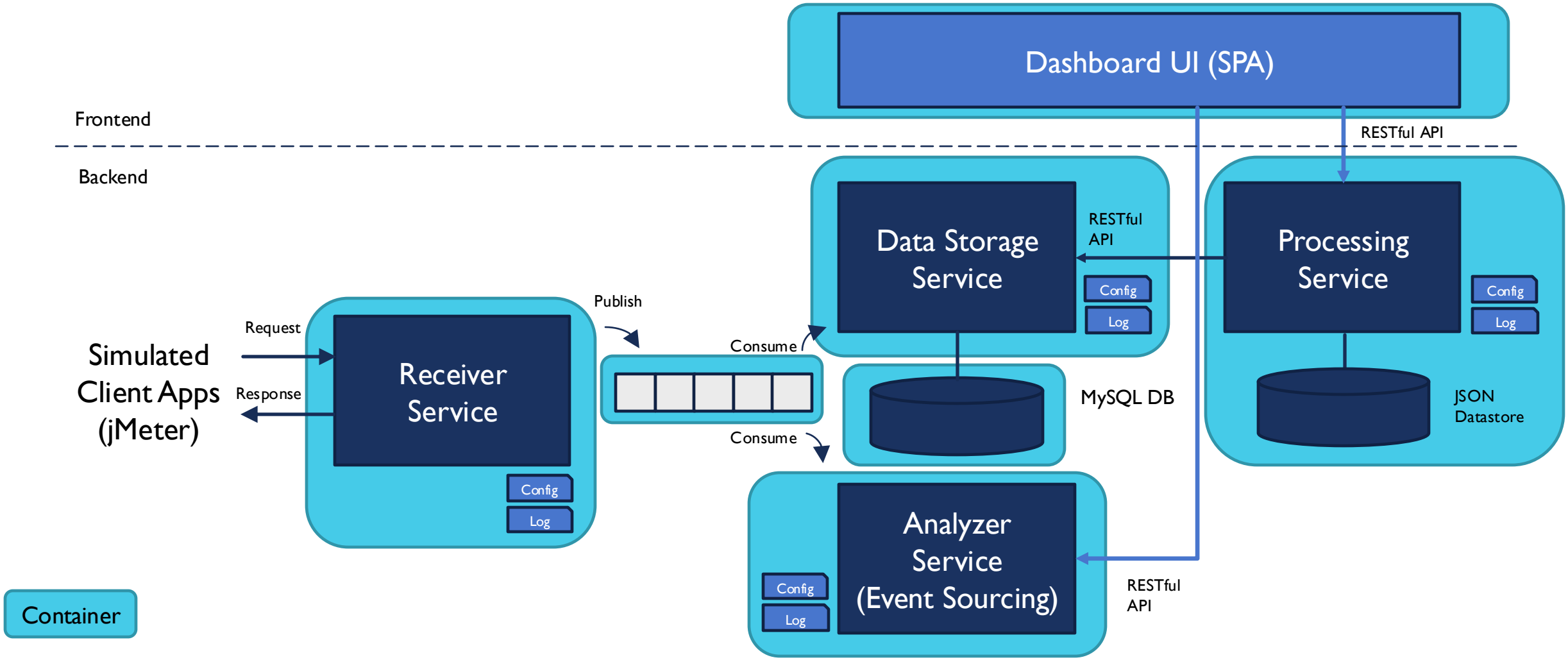
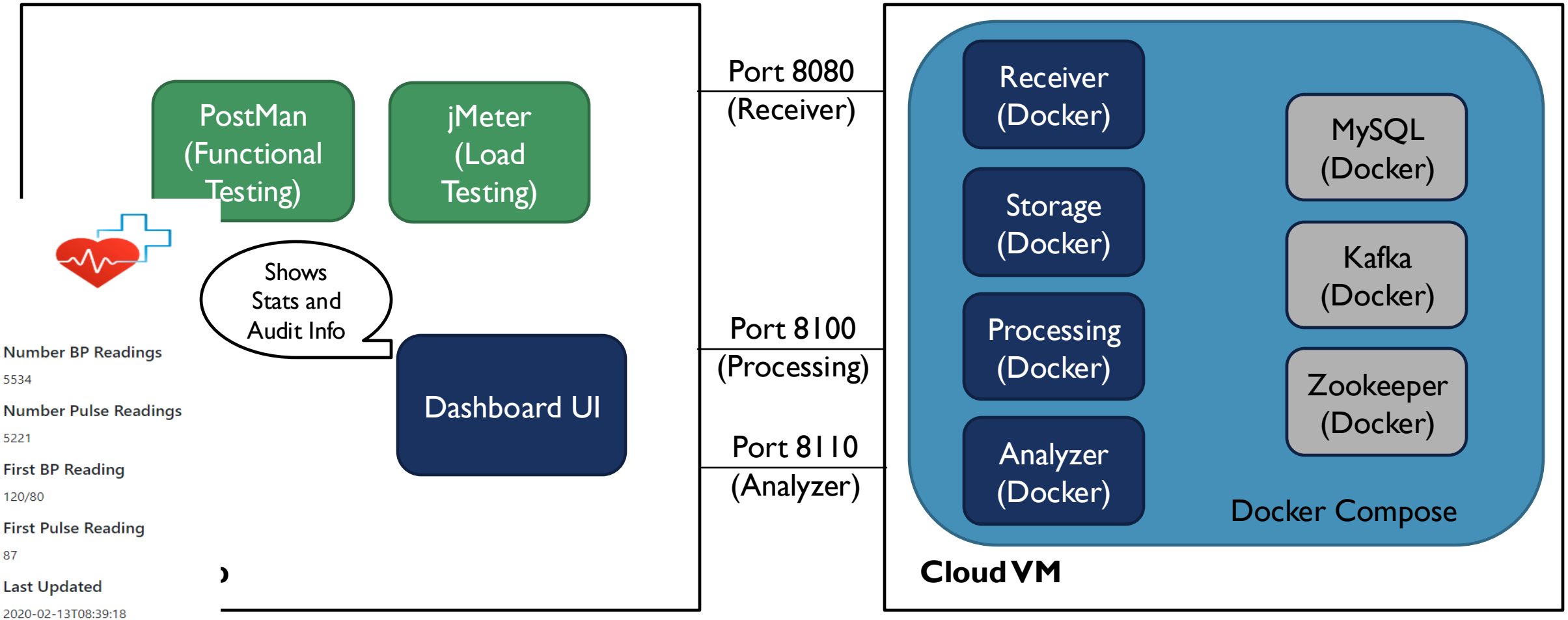


# LAB 10 TEST ENVIRONMENT



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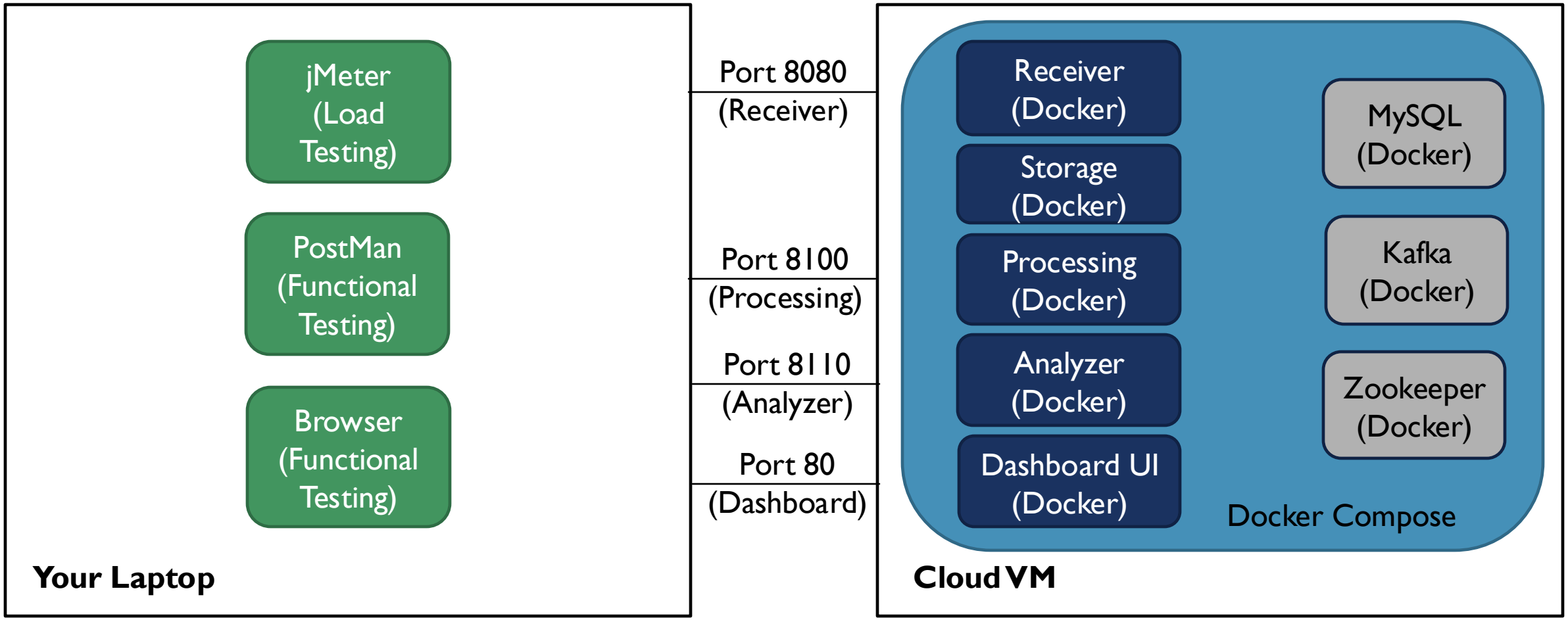

# DASHBOARD UI

- Lets go through the updates you will need to make in the sample code…
- You can use a pure HTML/JS implementation

# TODAY'S LAB

Today you will:

- Demo your Lab 9 by the end of class.
- Work on Lab 10 – It is due for Demo next class.